QS30-6 REV.A　系统框图 / 电源树 / 安全使能链

粗线=功率路径，细线=信号；标号为原理图网络名。所有门控在硬件中建立默认关断状态。

J1 12 V DC 输入
F1 PPTC 1.5 A / D1 SMBJ16A TVS / D2 SS54 反接保护

EXT12

EXT12 保护后 12 V
供：5V buck / DRV_VDD 开关 / 升压开关

U1 TPS54202 12V→5.08V/2A
L1 10µH · C6 22µF · EN UVLO≈4.8 V

D3/D4 双肖特基 OR
EXT5 / VBUS → 5V_SYS，反向阻断，无回灌

3V3

U2 SY8089A 5V→3.315V/2A
L2 2.2µH · C11 22µF · 3.3 V 预算 800 mA

U10 ESP32-S3-WROOM-1-N8
GPIO11/12/14 = DIN/CLK/LAT · GPIO15 = DISP_ALLOW · GPIO16 = HV_EN_REQ · GPIO1 = HV_SENSE · GPIO7 = HV_FAULT_N · GPIO19/20 = USB D-/D+ · GPIO43/44 = UART0 · 天线端伸出板边 + 禁布区

VBUS

J2 USB-C 16P (2.0 device)
CC1/CC2 5.1 kΩ · U11 USBLC6-2SC6 ESD · 22 Ω 串联 · VBUS 仅逻辑下载

DISP_DIN/CLK/LAT/ALLOW

HV_EN_REQ

Q1 DRV_VDD 高边开关
默认关断（栅极 100 kΩ 上拉到 EXT12）
DRV_VDD ≈ 11.5 V → HV5622 VDD

DRV_VDD_EN

电源监控 + 硬件使能门控
U5/U8 LM393 + U6 TL431(2.5 V) + U7 74HC08
EXT12_PG(11.0 V) · 3V3_PG(3.11 V) · DRV_PG(10.41 V) · FAULT_OK
DRV_VDD_EN = EXT12_PG·3V3_PG·HV_EN_REQ
HV_EN = DRV_PG·FAULT_OK·HV_EN_REQ

U3/U4 CD40109B 电平转换
3.3 V → 12 V：DIN/CLK/LAT/ALLOW + HV_EN_REQ
VCC=3V3, VDD=DRV_VDD；EN=3V3，3V3 失电即高阻
12 V 域全部 10 kΩ 下拉（BL 低=消隐）

DRV_*

U21/U22 HV5622 (PQFP-44) x2
32+32=64 路开漏，60 路用于 6×10 数字阴极
VDD=DRV_VDD · POL≡DRV_VDD · BL 低=全关
DATA 级联：U21→U22，帧长 64 bit

EXT12

HV_EN

DRV_PG/FAULT_OK

12 V → 172 V 升压 (U20 UC3843)
fsw≈78 kHz · L20 220 µH (Isat≥5 A) · Q20 IRF740 · D20 UF4007
C50/C51 20 µF/250 V · R67/R68 200 kΩ 泄放（172→50 V≈3.8 s）
HV_EN → Q26/Q27 高边开关（默认关断）

独立硬件过压保护 / 锁存
U23 LM393 + U24 TL431 · 门限 198.8 V (2.5 V·1519.1/19.1)
OVP_SET → Q21/Q22 分立 SCR 锁存 → Q23 拉低 COMP，Q24 拉低 HV_FAULT_N；Q25 由 GPIO18 受控复位

HV_BUS 172 V
C50/C51/C52 · 反馈 2×750 k/22.1 k
R27/R28 1 MΩ×2 分压 → ADC1_CH0（1.70 V @172 V）

172 V

六路阳极限流
每管 2×10 kΩ/2 W 串联 = 20 kΩ
正常工作 ≈2.35 mA；单管短路耗散 ≈1.45 W

J31..J36 QS30-1 管座
13 脚 1:1 映射；1/8 脚不接
阴极 → HV5622 HVOUT（60 路）

注：DRV_VDD / HV_BUS / 阴极网络按 HV 或 DRV 网络类别设置 PCB 规则；天线禁布区与高压区间距见预布局图（H07/H08 未关闭）。

帧长由 96 bit（3 片）修正为 64 bit（2 片，32×2），详见 REV_A_CHANGELOG.md。

QS30-6 REV.A　01 电源输入与保护 / 12V-5V-3.3V / DRV_VDD 负载开关

连接表式原理图：每引脚列出 引脚号/名称 → 网络。网络颜色 红=HV 橙=DRV_VDD域 蓝=电源 紫=USB差分 绿=信号 灰=GND。与 CSV 网表同源生成。

J1 DC-005-5.08-2P
[TERM-2] DC12 IN 5.08mm 2P PROVISIONAL

1	1	EXT12_RAW
2	2	GND

D2 SS54
[SMB] 40 V 5 A 肖特基 VERIFIED

1	K	EXT12
2	A	EXT12_F

C3 100nF/50V
[0603] 0.1 µF 50 V X7R VERIFIED

1	1	EXT12
2	2	GND

R2 33k
[0603] 33 kΩ 1% VERIFIED

1	1	EN_U1
2	2	GND

L1 10uH/3A
[0630] 10 µH 3 A 屏蔽功率电感 PROVISIONAL

1	1	SW_U1
2	2	EXT5

C6 22uF/10V
[0805] 22 µF 10 V X5R VERIFIED

1	1	EXT5
2	2	GND

D4 SS14
[SMA] 40 V 1 A 肖特基 VERIFIED

1	K	5V_SYS
2	A	VBUS

U2 SY8089AAAC
[SOT-23-5] SY8089A 5V->3.3V 2A VERIFIED

1	EN	EN_U2
2	GND	GND
3	LX	SW_U2
4	IN	5V_SYS
5	FB	FB_U2

R6 100k
[0603] 100 kΩ 1% VERIFIED

1	1	3V3
2	2	FB_U2

C12 100nF/16V
[0603] 0.1 µF 16 V X7R VERIFIED

1	1	3V3
2	2	GND

Q2 2N7002
[SOT-23] N-MOSFET 60 V VERIFIED

1	G	Q2_G
2	S	GND
3	D	Q1_G

C13 10uF/25V
[1206] 10 µF 25 V X5R VERIFIED

1	1	DRV_VDD
2	2	GND

F1 PPTC-1.5A-1812
[1812] 1.5 A 自恢复保险丝 PROVISIONAL

1	1	EXT12_RAW
2	2	EXT12_F

C1 100uF/25V
[8x10.5 SMD] 100 µF 25 V 电解 VERIFIED

1	1	EXT12
2	2	GND

U1 TPS54202DDCR
[SOT-23-6] TPS54202 12V->5.0V 2A VERIFIED

1	GND	GND
2	SW	SW_U1
3	VIN	EXT12
4	FB	FB_U1
5	EN	EN_U1
6	BOOT	BOOT_U1

C5 100nF/50V
[0603] 0.1 µF 50 V X7R VERIFIED

1	1	BOOT_U1
2	2	SW_U1

R4 13.3k
[0603] 13.3 kΩ 1% VERIFIED

1	1	FB_U1
2	2	GND

D3 SS14
[SMA] 40 V 1 A 肖特基 VERIFIED

1	K	5V_SYS
2	A	EXT5

C9 100nF/16V
[0603] 0.1 µF 16 V X7R VERIFIED

1	1	5V_SYS
2	2	GND

C10 10uF/10V
[0805] 10 µF 10 V X5R VERIFIED

1	1	5V_SYS
2	2	GND

R7 22.1k
[0603] 22.1 kΩ 1% VERIFIED

1	1	FB_U2
2	2	GND

Q1 DMP3098L
[SOT-23] P-MOSFET 30 V/3.8 A PROVISIONAL

1	G	Q1_G
2	S	EXT12
3	D	DRV_VDD

R9 100Ω
[0603] 100 Ω 1% VERIFIED

1	1	DRV_VDD_EN
2	2	Q2_G

C14 100nF/50V
[0603] 0.1 µF 50 V X7R VERIFIED

1	1	DRV_VDD
2	2	GND

D1 SMBJ16A
[SMB] TVS 16 V 单向 PROVISIONAL

1	K	EXT12_F
2	A	GND

C2 1uF/50V
[1206] 1 µF 50 V X7R VERIFIED

1	1	EXT12
2	2	GND

R1 100k
[0603] 100 kΩ 1% VERIFIED

1	1	EXT12
2	2	EN_U1

C4 10uF/25V
[1206] 10 µF 25 V X5R VERIFIED

1	1	EXT12
2	2	GND

R3 100k
[0603] 100 kΩ 1% VERIFIED

1	1	EXT5
2	2	FB_U1

C7 100nF/16V
[0603] 0.1 µF 16 V X7R VERIFIED

1	1	EXT5
2	2	GND

C8 22uF/10V
[0805] 22 µF 10 V X5R VERIFIED

1	1	5V_SYS
2	2	GND

R5 100k
[0603] 100 kΩ 1% VERIFIED

1	1	5V_SYS
2	2	EN_U2

L2 2.2uH/3A
[0520] 2.2 µH 3 A 屏蔽功率电感 PROVISIONAL

1	1	SW_U2
2	2	3V3

C11 22uF/6.3V
[0805] 22 µF 6.3 V X5R VERIFIED

1	1	3V3
2	2	GND

R8 100k
[0603] 100 kΩ 1% VERIFIED

1	1	EXT12
2	2	Q1_G

R10 100k
[0603] 100 kΩ 1% VERIFIED

1	1	Q2_G
2	2	GND

QS30-6 REV.A　02 ESP32-S3-WROOM-1-N8 与 USB-C / UART0 / 按键接口

连接表式原理图：每引脚列出 引脚号/名称 → 网络。网络颜色 红=HV 橙=DRV_VDD域 蓝=电源 紫=USB差分 绿=信号 灰=GND。与 CSV 网表同源生成。

U10 ESP32-S3-WROOM-1-N8

[WROOM-1 41pad] ESP32-S3 8MB Flash 无 PSRAM VERIFIED

1	GND	GND	15	IO3	NC	29	IO36	NC
2	3V3	3V3	16	IO46	GPIO46_STRAP	30	IO37	NC
3	EN	EN_MCU	17	IO9	I2C_SCL	31	IO38	NC
4	IO4	KEY_SET	18	IO10	NC	32	IO39	NC
5	IO5	KEY_PLUS	19	IO11	DISP_DIN	33	IO40	NC
6	IO6	KEY_MINUS	20	IO12	DISP_CLK	34	IO41	NC
7	IO7	HVF_NODE	21	IO13	EXP_IO13	35	IO42	NC
8	IO15	DISP_ALLOW	22	IO14	DISP_LAT	36	IO44/RXD0	UART0_RX
9	IO16	HV_EN_REQ	23	IO21	EXP_IO21	37	IO43/TXD0	UART0_TX
10	IO17	STATUS_LED	24	IO47	NC	38	IO2	NC
11	IO18	OVP_RST	25	IO48	NC	39	IO1	HV_SENSE_ADC
12	IO8	I2C_SDA	26	IO45	NC	40	GND	GND
13	IO19	USB_DM_MCU	27	IO0	BOOT_N	41	GND/EP	GND
14	IO20	USB_DP_MCU	28	IO35	NC			

C16 22uF

[0805] 22uF 6.3V VERIFIED

1	1	3V3
2	2	GND

C17 1uF

[0603] 1uF 16V VERIFIED

1	1	3V3
2	2	GND

C18 100nF

[0603] 100nF 16V VERIFIED

1	1	3V3
2	2	GND

R11 10k

[0603] 10 kΩ 1% VERIFIED

1	1	3V3
2	2	EN_MCU

C19 1uF/16V

[0603] 1 µF 16 V X7R VERIFIED

1	1	EN_MCU
2	2	GND

SW1 TS-1187A

[4P 6x6 SMD] 复位按键 PROVISIONAL

1	1	EN_MCU
2	2	GND

SW2 TS-1187A

[4P 6x6 SMD] BOOT 按键 PROVISIONAL

1	1	BOOT_N
2	2	GND

R12 10k

[0603] 10 kΩ 1% VERIFIED

1	1	3V3
2	2	BOOT_N

R13 10k

[0603] 10 kΩ 1% VERIFIED

1	1	GPIO46_STRAP
2	2	GND

J2 USB-C-16P-2.0

[SMD 16P] USB-C 母座 16P USB2.0 PROVISIONAL

A1	GND	GND	B5	CC2	CC2			
A4	VBUS	VBUS	B6	D+	USB_DP_CON			
A5	CC1	CC1	B7	D-	USB_DM_CON			
A6	D+	USB_DP_CON	B8	SBU2	NC			
A7	D-	USB_DM_CON	B9	VBUS	VBUS			
A8	SBU1	NC	B12	GND	GND			
A9	VBUS	VBUS	S1	SHIELD	SHIELD			
A12	GND	GND	S2	SHIELD	SHIELD			
B1	GND	GND	S3	SHIELD	SHIELD			
B4	VBUS	VBUS	S4	SHIELD	SHIELD			

R14 5.1k

[0603] 5.1 kΩ 1% VERIFIED

1	1	CC1
2	2	GND

R15 5.1k

[0603] 5.1 kΩ 1% VERIFIED

1	1	CC2
2	2	GND

R16 1M

[0603] 1 MΩ 1% VERIFIED

1	1	SHIELD
2	2	GND

C20 4.7nF/50V

[0603] 4.7 nF 50 V X7R VERIFIED

1	1	SHIELD
2	2	GND

U11 USBLC6-2SC6

[SOT-23-6] USB ESD 保护 VERIFIED

1	I/O1	USB_DP_CON
2	GND	GND
3	I/O2	USB_DM_CON
4	I/O2	USB_DM_ESD
5	VBUS	VBUS
6	I/O1	USB_DP_ESD

C21 1uF/16V

[0603] 1 µF 16 V X7R VERIFIED

1	1	VBUS
2	2	GND

C22 100nF/16V

[0603] 0.1 µF 16 V X7R VERIFIED

1	1	VBUS
2	2	GND

J3 HDR-1x6-2.54

[2.54mm] UART0 拔插排针 1x6 VERIFIED

1	1	GND
2	2	UART_3V3_REF
3	3	UART0_TX_H
4	4	UART0_RX
5	5	EN_MCU
6	6	BOOT_N

R17 22Ω

[0402] 22 Ω 1% PROVISIONAL

1	1	USB_DP_ESD
2	2	USB_DP_MCU

R18 22Ω

[0402] 22 Ω 1% PROVISIONAL

1	1	USB_DM_ESD
2	2	USB_DM_MCU

C23 10pF/50V　[DNP]

[0402] 10 pF 50 V PROVISIONAL

1	1	USB_DP_MCU
2	2	GND

C24 10pF/50V　[DNP]

[0402] 10 pF 50 V PROVISIONAL

1	1	USB_DM_MCU
2	2	GND

R19 499Ω

[0603] 499 Ω 1% VERIFIED

1	1	UART0_TX
2	2	UART0_TX_H

R20 0Ω　[DNP]

[0603] 0 Ω VERIFIED

1	1	3V3
2	2	UART_3V3_REF

SW3 TS-1187A

[4P 6x6 SMD] SET 按键 PROVISIONAL

1	1	KEY_SET
2	2	GND

SW4 TS-1187A

[4P 6x6 SMD] PLUS 按键 PROVISIONAL

1	1	KEY_PLUS
2	2	GND

SW5 TS-1187A

[4P 6x6 SMD] MINUS 按键 PROVISIONAL

1	1	KEY_MINUS
2	2	GND

R21 10k

[0603] 10 kΩ 1% VERIFIED

1	1	KEY_SET
2	2	3V3

R22 10k

[0603] 10 kΩ 1% VERIFIED

1	1	KEY_PLUS
2	2	3V3

R23 10k

[0603] 10 kΩ 1% VERIFIED

1	1	KEY_MINUS
2	2	3V3

C25 100nF/16V

[0603] 0.1 µF 16 V X7R VERIFIED

1	1	KEY_SET
2	2	GND

C26 100nF/16V

[0603] 0.1 µF 16 V X7R VERIFIED

1	1	KEY_PLUS
2	2	GND

C27 100nF/16V

[0603] 0.1 µF 16 V X7R VERIFIED

1	1	KEY_MINUS
2	2	GND

02 ESP32-S3-WROOM-1-N8 与 USB-C / UART0 / 按键接口（续）

LED1 LTST-C190KRKT
[0603] 红色 LED VERIFIED
1 K GND
2 A LED1_A

R24 1k
[0603] 1 kΩ 1% VERIFIED
1 1 STATUS_LED
2 2 LED1_A

R25 4.7k [DNP]
[0603] 4.7 kΩ 1% VERIFIED
1 1 3V3
2 2 I2C_SDA

R26 4.7k [DNP]
[0603] 4.7 kΩ 1% VERIFIED
1 1 3V3
2 2 I2C_SCL

R27 1M
[1206] 1 MΩ 1% 1206 PROVISIONAL
1 1 HV_BUS
2 2 HVS_MID

R28 1M
[1206] 1 MΩ 1% 1206 PROVISIONAL
1 1 HVS_MID
2 2 HVS_NODE

R29 20k
[0805] 20 kΩ 1% PROVISIONAL
1 1 HVS_NODE
2 2 GND

R30 1k
[0603] 1 kΩ 1% PROVISIONAL
1 1 HVS_NODE
2 2 HV_SENSE_ADC

C28 100nF/16V
[0603] 0.1 µF 16 V X7R VERIFIED
1 1 HV_SENSE_ADC
2 2 GND

D5 BAT54
[SOD-323] 肖特基 30 V VERIFIED
1 K 3V3
2 A HV_SENSE_ADC

D6 BAT54
[SOD-323] 肖特基 30 V VERIFIED
1 K HV_SENSE_ADC
2 A GND

R31 100k
[0603] 100 kΩ 1% PROVISIONAL
1 1 HV_FAULT_OD
2 2 HVF_NODE

R32 36.5k
[0603] 36.5 kΩ 1% PROVISIONAL
1 1 HVF_NODE
2 2 GND

C29 100nF/16V
[0603] 0.1 µF 16 V VERIFIED
1 1 HVF_NODE
2 2 GND

D7 BAT54
[SOD-323] 肖特基 30 V VERIFIED
1 K 3V3
2 A HVF_NODE

D8 BAT54
[SOD-323] 肖特基 30 V VERIFIED
1 K HVF_NODE
2 A GND

QS30-6 REV.A　03 HV5622 高压驱动 x2 / 3.3V→DRV_VDD 电平转换 / 监控与门控

连接表式原理图：每引脚列出 引脚号/名称 → 网络。网络颜色 红=HV 橙=DRV_VDD域 蓝=电源 紫=USB差分 绿=信号 灰=GND。与 CSV 网表同源生成。

U3 CD40109B
[SOIC-16] 四路低→高电平转换 #1 VERIFIED

1	VCC	3V3	9	ENABLE C	3V3
2	ENABLE A	3V3	10	C	DISP_LAT
3	A	DISP_DIN	11	G	DRV_LAT
4	E	DRV_DIN	12	NC	NC
5	F	DRV_CLK	13	H	DRV_BL
6	B	DISP_CLK	14	D	DISP_ALLOW
7	ENABLE B	3V3	15	ENABLE D	3V3
8	VSS	GND	16	VDD	DRV_VDD

R34 10k
[0603] 10k 1% VERIFIED

1	1	DRV_CLK
2	2	GND

R37 10k
[0603] 10k 1% VERIFIED

1	1	DRV_HV_EN_REQ
2	2	GND

U6 TL431
[SOT-23] 2.5 V 并联基准 VERIFIED

1	CATHODE	VREF_2V5
2	REF	VREF_2V5
3	ANODE	GND

C32 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	VREF_2V5
2	2	GND

R39 100k
[0603] 100 kΩ 1% VERIFIED

1	1	EXT12
2	2	EXT12_SNS

R41 10k
[0603] 10 kΩ 1% VERIFIED

1	1	EXT12_PG
2	2	3V3

C36 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	P3V3_SNS
2	2	GND

R46 31.6k
[0603] 31.6 kΩ 1% VERIFIED

1	1	DRV_SNS
2	2	GND

R48 10k
[0603] 10 kΩ 1% VERIFIED

1	1	FAULT_OK
2	2	3V3

R49 100k
[0603] 100 kΩ 1% VERIFIED

1	1	HV_EN
2	2	GND

R52 100k
[0603] 100 kΩ 1% VERIFIED

1	1	DISP_ALLOW
2	2	GND

U4 CD40109B
[SOIC-16] 四路低→高电平转换 #2 VERIFIED

1	VCC	3V3	9	ENABLE C	3V3
2	ENABLE A	3V3	10	C	GND
3	A	HV_EN_REQ	11	G	NC
4	E	DRV_HV_EN_REQ	12	NC	NC
5	F	NC	13	H	NC
6	B	GND	14	D	GND
7	ENABLE B	3V3	15	ENABLE D	3V3
8	VSS	GND	16	VDD	DRV_VDD

R35 10k
[0603] 10k 1% VERIFIED

1	1	DRV_LAT
2	2	GND

U5 LM393DR
[SOIC-8] 双比较器 #1（EXT12 / 3V3 监控） VERIFIED

1	OUT1	EXT12_PG
2	IN1-	VREF_2V5
3	IN1+	EXT12_SNS
4	GND	GND
5	IN2+	P3V3_SNS
6	IN2-	VREF_2V5
7	OUT2	P3V3_PG
8	VCC	3V3

R38 680Ω
[0603] 680 Ω 1% VERIFIED

1	1	3V3
2	2	VREF_2V5

C34 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	3V3
2	2	GND

C35 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	EXT12_SNS
2	2	GND

R43 41.2k
[0603] 41.2 kΩ 1% VERIFIED

1	1	P3V3_SNS
2	2	GND

R45 100k
[0603] 100 kΩ 1% VERIFIED

1	1	DRV_VDD
2	2	DRV_SNS

R47 10k
[0603] 10 kΩ 1% VERIFIED

1	1	DRV_PG
2	2	3V3

C38 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	3V3
2	2	GND

R51 100k
[0603] 100 kΩ 1% VERIFIED

1	1	HV_EN_REQ
2	2	GND

C30 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	3V3
2	2	GND

C31 100nF/50V
[0603] 0.1 µF 50 V VERIFIED

1	1	DRV_VDD
2	2	GND

R33 10k
[0603] 10k 1% VERIFIED

1	1	DRV_DIN
2	2	GND

R36 10k
[0603] 10k 1% VERIFIED

1	1	DRV_BL
2	2	GND

U8 LM393DR
[SOIC-8] 双比较器 #2（DRV_VDD / 故障状态） VERIFIED

1	OUT1	DRV_PG
2	IN1-	VREF_2V5
3	IN1+	DRV_SNS
4	GND	GND
5	IN2+	HVF_NODE
6	IN2-	VREF_2V5
7	OUT2	FAULT_OK
8	VCC	3V3

C33 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	3V3
2	2	GND

R40 30.1k
[0603] 30.1 kΩ 1% VERIFIED

1	1	EXT12_SNS
2	2	GND

R42 10k
[0603] 10 kΩ 1% VERIFIED

1	1	3V3
2	2	P3V3_SNS

R44 10k
[0603] 10 kΩ 1% VERIFIED

1	1	P3V3_PG
2	2	3V3

C37 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	DRV_SNS
2	2	GND

U7 74HC08
[SOIC-14] 四路与门 VERIFIED

1	1A	EXT12_PG	8	3Y	DRV_OK
2	1B	P3V3_PG	9	3A	DRV_PG
3	1Y	PG_12V3V	10	3B	FAULT_OK
4	2A	PG_12V3V	11	4Y	HV_EN
5	2B	HV_EN_REQ	12	4A	DRV_OK
6	2Y	DRV_VDD_EN	13	4B	HV_EN_REQ
7	GND	GND	14	VCC	3V3

03 HV5622 高压驱动 x2 / 3.3V→DRV_VDD 电平转换 / 监控与门控（续）

U21 HV5622PJ

[PQFP-44] HV5622 32 路开漏高压驱动 VERIFIED

1	HVOUT22	K_T3_1	16	HVOUT7	K_T1_6	31	LE DRV_LAT
2	HVOUT21	K_T3_0	17	HVOUT6	K_T1_5	32	DATA IN DRV_DIN
3	HVOUT20	K_T2_9	18	HVOUT5	K_T1_4	33	BL DRV_BL
4	HVOUT19	K_T2_8	19	HVOUT4	K_T1_3	34	NC NC
5	HVOUT18	K_T2_7	20	HVOUT3	K_T1_2	35	HVOUT32 K_T4_1
6	HVOUT17	K_T2_6	21	HVOUT2	K_T1_1	36	HVOUT31 K_T4_0
7	HVOUT16	K_T2_5	22	HVOUT1	K_T1_0	37	HVOUT30 K_T3_9
8	HVOUT15	K_T2_4	23	DATA OUT	DRV1_DOUT	38	HVOUT29 K_T3_8
9	HVOUT14	K_T2_3	24	NC	NC	39	HVOUT28 K_T3_7
10	HVOUT13	K_T2_2	25	NC	NC	40	HVOUT27 K_T3_6
11	HVOUT12	K_T2_1	26	NC	NC	41	HVOUT26 K_T3_5
12	HVOUT11	K_T2_0	27	POL	DRV_VDD	42	HVOUT25 K_T3_4
13	HVOUT10	K_T1_9	28	CLK	DRV_CLK	43	HVOUT24 K_T3_3
14	HVOUT9	K_T1_8	29	VSS	GND	44	HVOUT23 K_T3_2
15	HVOUT8	K_T1_7	30	VDD	DRV_VDD		

C39 10uF/25V

[1206] 10 µF 25 V X5R VERIFIED

1	1	DRV_VDD
2	2	GND

C41 100nF/50V

[0603] 0.1 µF 50 V X7R VERIFIED

1	1	DRV_VDD
2	2	GND

U22 HV5622PJ

[PQFP-44] HV5622 32 路开漏高压驱动 VERIFIED

1	HVOUT22	K_T6_3	16	HVOUT7	K_T4_8	31	LE DRV_LAT
2	HVOUT21	K_T6_2	17	HVOUT6	K_T4_7	32	DATA IN DRV1_DOUT
3	HVOUT20	K_T6_1	18	HVOUT5	K_T4_6	33	BL DRV_BL
4	HVOUT19	K_T6_0	19	HVOUT4	K_T4_5	34	NC NC
5	HVOUT18	K_T5_9	20	HVOUT3	K_T4_4	35	HVOUT32 NC
6	HVOUT17	K_T5_8	21	HVOUT2	K_T4_3	36	HVOUT31 NC
7	HVOUT16	K_T5_7	22	HVOUT1	K_T4_2	37	HVOUT30 NC
8	HVOUT15	K_T5_6	23	DATA OUT	DRV2_DOUT	38	HVOUT29 NC
9	HVOUT14	K_T5_5	24	NC	NC	39	HVOUT28 K_T6_9
10	HVOUT13	K_T5_4	25	NC	NC	40	HVOUT27 K_T6_8
11	HVOUT12	K_T5_3	26	NC	NC	41	HVOUT26 K_T6_7
12	HVOUT11	K_T5_2	27	POL	DRV_VDD	42	HVOUT25 K_T6_6
13	HVOUT10	K_T5_1	28	CLK	DRV_CLK	43	HVOUT24 K_T6_5
14	HVOUT9	K_T5_0	29	VSS	GND	44	HVOUT23 K_T6_4
15	HVOUT8	K_T4_9	30	VDD	DRV_VDD		

C40 10uF/25V

[1206] 10 µF 25 V X5R VERIFIED

1	1	DRV_VDD
2	2	GND

C42 100nF/50V

[0603] 0.1 µF 50 V X7R VERIFIED

1	1	DRV_VDD
2	2	GND

QS30-6 REV.A　04 12V→172V 升压 / 独立硬件过压保护与锁存

连接表式原理图：每引脚列出 引脚号/名称 → 网络。网络颜色 红=HV 橙=DRV_VDD域 蓝=电源 紫=USB差分 绿=信号 灰=GND。与 CSV 网表同源生成。

Q26 DMP3098L
[SOT-23] P-MOSFET 30 V/3.8 A PROVISIONAL

1	G	Q26_G
2	S	EXT12
3	D	HV_IN

C43 10uF/25V
[1206] 10 µF 25 V X5R VERIFIED

1	1	HV_IN
2	2	GND

R60 10Ω
[0805] 10 Ω 1% VERIFIED

1	1	HV_IN
2	2	VCC_HV

C45 100nF/25V
[0603] 0.1 µF 25 V VERIFIED

1	1	VCC_HV
2	2	GND

C47 2.2nF/50V
[0603] 2.2 nF 50 V VERIFIED

1	1	RCT_N
2	2	GND

C48 470pF/50V
[0603] 470 pF 50 V VERIFIED

1	1	CS_FILT
2	2	GND

Q20 IRF740
[TO-220] N-MOSFET 400 V/10 A VERIFIED

1	G	Q20_G
2	D	SW_HV
3	S	CS_N

C49 10nF/630V
[1206] 10 nF 630 V X7R VERIFIED

1	1	CLAMP_N
2	2	HV_BUS

C51 10uF/250V
[10x12.5 SMD] 10 µF 250 V 电解 PROVISIONAL

1	1	HV_BUS
2	2	GND

R68 100k
[1206] 100 kΩ 1% VERIFIED

1	1	BLEED_MID
2	2	GND

R70 750k
[1206] 750 kΩ 1% VERIFIED

1	1	FB_MID
2	2	FB_HV

C55 22uF/16V
[0805] 22 µF 16 V X5R VERIFIED

1	1	COMP_N
2	2	GND

U23 LM393DR
[SOIC-8] OVP 比较器 VERIFIED

1	OUT1	OVP_SET
2	IN1-	VREF_OVP
3	IN1+	OVP_SNS
4	GND	GND
5	IN2+	NC
6	IN2-	NC
7	OUT2	NC
8	VCC	EXT12

R87 100k
[0603] 100 kΩ 1% VERIFIED

1	1	EXT12
2	2	Q26_G

Q27 2N7002
[SOT-23] N-MOSFET 60 V VERIFIED

1	G	HV_EN_G
2	S	GND
3	D	Q26_G

C44 10uF/25V
[1206] 10 µF 25 V X5R VERIFIED

1	1	VCC_HV
2	2	GND

C46 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	VREF5_HV
2	2	GND

R62 1Ω 1W
[2512] 1 Ω 1 W 合金采样 PROVISIONAL

1	1	CS_N
2	2	GND

R64 22Ω
[0603] 22 Ω 1% VERIFIED

1	1	GATE_N
2	2	Q20_G

L20 220uH/5A
[12x12 屏蔽] 220 µH ≥5 A 功率电感（Isat≥5A） PROVISIONAL

1	1	HV_IN
2	2	SW_HV

D21 UF4007
[SMA/DO-41] 1000 V 1 A 快恢复 VERIFIED

1	K	CLAMP_N
2	A	SW_HV

C50 10uF/250V
[10x12.5 SMD] 10 µF 250 V 电解 PROVISIONAL

1	1	HV_BUS
2	2	GND

R67 100k
[1206] 100 kΩ 1% VERIFIED

1	1	HV_BUS
2	2	BLEED_MID

R69 750k
[1206] 750 kΩ 1% VERIFIED

1	1	HV_BUS
2	2	FB_MID

C54 100nF/50V
[0603] 0.1 µF 50 V VERIFIED

1	1	FB_HV
2	2	GND

C56 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	COMP_R
2	2	GND

R74 4.7k
[0603] 4.7 kΩ 1% VERIFIED

1	1	EXT12
2	2	VREF_OVP

C57 100nF/16V
[0603] 0.1 µF 16 V VERIFIED

1	1	VREF_OVP
2	2	GND

R88 100Ω
[0603] 100 Ω 1% VERIFIED

1	1	HV_EN
2	2	HV_EN_G

R89 100k
[0603] 100 kΩ 1% VERIFIED

1	1	HV_EN_G
2	2	GND

U20 UC3843AN
[DIP-8/SOIC-8] UC3843 电流型 PWM 控制器 VERIFIED

1	COMP	COMP_N
2	FB	FB_HV
3	CS	CS_FILT
4	RT/CT	RCT_N
5	GND	GND
6	OUT	GATE_N
7	VCC	VCC_HV
8	VREF	VREF5_HV

R61 10k
[0603] 10 kΩ 1% VERIFIED

1	1	VREF5_HV
2	2	RCT_N

R63 100Ω
[0603] 100 Ω 1% VERIFIED

1	1	CS_N
2	2	CS_FILT

R65 10k
[0603] 10 kΩ 1% VERIFIED

1	1	Q20_G
2	2	GND

D20 UF4007
[SMA/DO-41] 1000 V 1 A 快恢复 VERIFIED

1	K	HV_BUS
2	A	SW_HV

R66 100k 2W
[2512] 100 kΩ 2 W PROVISIONAL

1	1	CLAMP_N
2	2	HV_BUS

C52 100nF/630V
[1206] 0.1 µF 630 V VERIFIED

1	1	HV_BUS
2	2	GND

C53 100nF/50V
[0603] 0.1 µF 50 V VERIFIED

1	1	BLEED_MID
2	2	GND

R71 22.1k
[0603] 22.1 kΩ 1% VERIFIED

1	1	FB_HV
2	2	GND

R72 10k
[0603] 10 kΩ 1% VERIFIED

1	1	COMP_N
2	2	COMP_R

U24 TL431
[SOT-23] 2.5 V 基准 VERIFIED

1	CATHODE	VREF_OVP
2	REF	VREF_OVP
3	ANODE	GND

R75 750k
[1206] 750 kΩ 1% VERIFIED

1	1	HV_BUS
2	2	OVP_MID

R76 750k
[1206] 750 kΩ 1% VERIFIED
1 1 OVP_MID
2 2 OVP_SNS

R78 10k
[0603] 10 kΩ 1% VERIFIED
1 1 OVP_SET
2 2 EXT12

Q22 MMBT3904
[SOT-23] NPN 40 V VERIFIED
1 B OVP_LATCH
2 E GND
3 C LATCH_B

C60 100nF/25V
[0603] 0.1 µF 25 V VERIFIED
1 1 OVP_LATCH
2 2 GND

Q24 MMBT3904
[SOT-23] NPN 40 V VERIFIED
1 B Q24_B
2 E GND
3 C HV_FAULT_OD

Q25 2N7002
[SOT-23] N-MOSFET 60 V VERIFIED
1 G OVP_RST_G
2 S GND
3 D OVP_LATCH

R77 19.1k
[0603] 19.1 kΩ 1% VERIFIED
1 1 OVP_SNS
2 2 GND

C59 100nF/25V
[0603] 0.1 µF 25 V VERIFIED
1 1 EXT12
2 2 GND

R79 100k
[0603] 100 kΩ 1% VERIFIED
1 1 LATCH_B
2 2 EXT12

R81 100k
[0603] 100 kΩ 1% VERIFIED
1 1 OVP_LATCH
2 2 GND

R82 47k
[0603] 47 kΩ 1% VERIFIED
1 1 OVP_LATCH
2 2 Q23_B

R84 100k
[0603] 100 kΩ 1% VERIFIED
1 1 HV_FAULT_OD
2 2 DRV_VDD

R86 100k
[0603] 100 kΩ 1% VERIFIED
1 1 OVP_RST_G
2 2 GND

C58 100nF/50V
[0603] 0.1 µF 50 V VERIFIED
1 1 OVP_SNS
2 2 GND

Q21 MMBT3906
[SOT-23] PNP 40 V VERIFIED
1 B LATCH_B
2 E EXT12
3 C OVP_LATCH

R80 10k
[0603] 10 kΩ 1% VERIFIED
1 1 OVP_SET
2 2 LATCH_B

Q23 MMBT3904
[SOT-23] NPN 40 V VERIFIED
1 B Q23_B
2 E GND
3 C COMP_N

R83 47k
[0603] 47 kΩ 1% VERIFIED
1 1 OVP_LATCH
2 2 Q24_B

R85 1k
[0603] 1 kΩ 1% VERIFIED
1 1 OVP_RST
2 2 OVP_RST_G

C61 100nF/16V
[0603] 0.1 µF 16 V VERIFIED
1 1 OVP_RST_G
2 2 GND

QS30-6 REV.A　05 六组 QS30-1 管座与阳极限流

连接表式原理图：每引脚列出 引脚号/名称 → 网络。网络颜色 红=HV 橙=DRV_VDD域 蓝=电源 紫=USB差分 绿=信号 灰=GND。与 CSV 网表同源生成。

J31 J13-1A-SOCKET

[13P socket] 13 脚可拆管座 (QS30-1) HOLD

1	IC_KEEP_OPEN	NC	8	IC8_KEEP_OPEN	NC
2	A	A_T1	9	K5	K_T1_5
3	K0	K_T1_0	10	K4	K_T1_4
4	K9	K_T1_9	11	K3	K_T1_3
5	K8	K_T1_8	12	K2	K_T1_2
6	K7	K_T1_7	13	K1	K_T1_1
7	K6	K_T1_6			

R90 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	HV_BUS
2	2	A_MID1

R91 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	A_MID1
2	2	A_T1

R92 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	HV_BUS
2	2	A_MID2

R93 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	A_MID2
2	2	A_T2

J32 J13-1A-SOCKET

[13P socket] 13 脚可拆管座 (QS30-1) HOLD

1	IC_KEEP_OPEN	NC	8	IC8_KEEP_OPEN	NC
2	A	A_T2	9	K5	K_T2_5
3	K0	K_T2_0	10	K4	K_T2_4
4	K9	K_T2_9	11	K3	K_T2_3
5	K8	K_T2_8	12	K2	K_T2_2
6	K7	K_T2_7	13	K1	K_T2_1
7	K6	K_T2_6			

J33 J13-1A-SOCKET

[13P socket] 13 脚可拆管座 (QS30-1) HOLD

1	IC_KEEP_OPEN	NC	8	IC8_KEEP_OPEN	NC
2	A	A_T3	9	K5	K_T3_5
3	K0	K_T3_0	10	K4	K_T3_4
4	K9	K_T3_9	11	K3	K_T3_3
5	K8	K_T3_8	12	K2	K_T3_2
6	K7	K_T3_7	13	K1	K_T3_1
7	K6	K_T3_6			

R94 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	HV_BUS
2	2	A_MID3

R95 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	A_MID3
2	2	A_T3

R96 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	HV_BUS
2	2	A_MID4

R97 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	A_MID4
2	2	A_T4

J34 J13-1A-SOCKET

[13P socket] 13 脚可拆管座 (QS30-1) HOLD

1	IC_KEEP_OPEN	NC	8	IC8_KEEP_OPEN	NC
2	A	A_T4	9	K5	K_T4_5
3	K0	K_T4_0	10	K4	K_T4_4
4	K9	K_T4_9	11	K3	K_T4_3
5	K8	K_T4_8	12	K2	K_T4_2
6	K7	K_T4_7	13	K1	K_T4_1
7	K6	K_T4_6			

J35 J13-1A-SOCKET

[13P socket] 13 脚可拆管座 (QS30-1) HOLD

1	IC_KEEP_OPEN	NC	8	IC8_KEEP_OPEN	NC
2	A	A_T5	9	K5	K_T5_5
3	K0	K_T5_0	10	K4	K_T5_4
4	K9	K_T5_9	11	K3	K_T5_3
5	K8	K_T5_8	12	K2	K_T5_2
6	K7	K_T5_7	13	K1	K_T5_1
7	K6	K_T5_6			

R98 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	HV_BUS
2	2	A_MID5

R99 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	A_MID5
2	2	A_T5

J36 J13-1A-SOCKET

[13P socket] 13 脚可拆管座 (QS30-1) HOLD

1	IC_KEEP_OPEN	NC	8	IC8_KEEP_OPEN	NC
2	A	A_T6	9	K5	K_T6_5
3	K0	K_T6_0	10	K4	K_T6_4
4	K9	K_T6_9	11	K3	K_T6_3
5	K8	K_T6_8	12	K2	K_T6_2
6	K7	K_T6_7	13	K1	K_T6_1
7	K6	K_T6_6			

R100 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	HV_BUS
2	2	A_MID6

R101 10k 2W

[2512] 10 kΩ 2 W 厚膜 PROVISIONAL

1	1	A_MID6
2	2	A_T6

TP1 TP-EXT12
[1.0mm pad] 测试点 EXT12 VERIFIED

1	1	EXT12

TP2 TP-EXT5
[1.0mm pad] 测试点 EXT5 VERIFIED

1	1	EXT5

TP3 TP-5V_SYS
[1.0mm pad] 测试点 5V_SYS VERIFIED

1	1	5V_SYS

TP4 TP-3V3
[1.0mm pad] 测试点 3V3 VERIFIED

1	1	3V3

TP5 TP-DRV_VDD
[1.0mm pad] 测试点 DRV_VDD VERIFIED

1	1	DRV_VDD

TP6 TP-HV_BUS
[1.0mm pad] 测试点 HV_BUS VERIFIED

1	1	HV_BUS

TP7 TP-GND
[1.0mm pad] 测试点 GND VERIFIED

1	1	GND

TP8 TP-EN_MCU
[1.0mm pad] 测试点 EN_MCU VERIFIED

1	1	EN_MCU

TP9 TP-BOOT_N
[1.0mm pad] 测试点 BOOT_N VERIFIED

1	1	BOOT_N

TP10 TP-UART0_TX
[1.0mm pad] 测试点 UART0_TX VERIFIED

1	1	UART0_TX

TP11 TP-UART0_RX
[1.0mm pad] 测试点 UART0_RX VERIFIED

1	1	UART0_RX

TP12 TP-HV_SENSE_ADC
[1.0mm pad] 测试点 HV_SENSE_ADC VERIFIED

1	1	HV_SENSE_ADC

TP13 TP-HVF_NODE
[1.0mm pad] 测试点 HVF_NODE VERIFIED

1	1	HVF_NODE

TP14 TP-EXT12_PG
[1.0mm pad] 测试点 EXT12_PG VERIFIED

1	1	EXT12_PG

TP15 TP-P3V3_PG
[1.0mm pad] 测试点 P3V3_PG VERIFIED

1	1	P3V3_PG

TP16 TP-DRV_PG
[1.0mm pad] 测试点 DRV_PG VERIFIED

1	1	DRV_PG

TP17 TP-HV_EN
[1.0mm pad] 测试点 HV_EN VERIFIED

1	1	HV_EN

TP18 TP-DRV_HV_EN_REQ
[1.0mm pad] 测试点 DRV_HV_EN_REQ VERIFIED

1	1	DRV_HV_EN_REQ

TP19 TP-OVP_LATCH
[1.0mm pad] 测试点 OVP_LATCH VERIFIED

1	1	OVP_LATCH

TP20 TP-DRV2_DOUT
[1.0mm pad] 测试点 DRV2_DOUT VERIFIED

1	1	DRV2_DOUT

J4 HDR-1x6-2.54
[2.54mm] 扩展接口 1x6 VERIFIED

1	1	3V3
2	2	GND
3	3	EXP_IO13
4	4	EXP_IO21
5	5	I2C_SDA
6	6	I2C_SCL

QS30-6 REV.A　60 路数字阴极通道映射表

管位→显示数字→QS30-1 物理脚号→驱动器输出→PQFP 引脚→串行位序。串行位序为假设约定，必须用 walking-one 实测确认。

管位	数字	QS30脚	阴极网络	逻辑通道	器件	输出	PQFP脚	串行位
J_T1	0	3	K_T1_0	0	U21	HVOUT1	22	63
J_T1	1	13	K_T1_1	1	U21	HVOUT2	21	62
J_T1	2	12	K_T1_2	2	U21	HVOUT3	20	61
J_T1	3	11	K_T1_3	3	U21	HVOUT4	19	60
J_T1	4	10	K_T1_4	4	U21	HVOUT5	18	59
J_T1	5	9	K_T1_5	5	U21	HVOUT6	17	58
J_T1	6	7	K_T1_6	6	U21	HVOUT7	16	57
J_T1	7	6	K_T1_7	7	U21	HVOUT8	15	56
J_T1	8	5	K_T1_8	8	U21	HVOUT9	14	55
J_T1	9	4	K_T1_9	9	U21	HVOUT10	13	54
J_T2	0	3	K_T2_0	10	U21	HVOUT11	12	53
J_T2	1	13	K_T2_1	11	U21	HVOUT12	11	52
J_T2	2	12	K_T2_2	12	U21	HVOUT13	10	51
J_T2	3	11	K_T2_3	13	U21	HVOUT14	9	50
J_T2	4	10	K_T2_4	14	U21	HVOUT15	8	49
J_T2	5	9	K_T2_5	15	U21	HVOUT16	7	48
J_T2	6	7	K_T2_6	16	U21	HVOUT17	6	47
J_T2	7	6	K_T2_7	17	U21	HVOUT18	5	46
J_T2	8	5	K_T2_8	18	U21	HVOUT19	4	45
J_T2	9	4	K_T2_9	19	U21	HVOUT20	3	44
J_T3	0	3	K_T3_0	20	U21	HVOUT21	2	43
J_T3	1	13	K_T3_1	21	U21	HVOUT22	1	42
J_T3	2	12	K_T3_2	22	U21	HVOUT23	44	41
J_T3	3	11	K_T3_3	23	U21	HVOUT24	43	40
J_T3	4	10	K_T3_4	24	U21	HVOUT25	42	39
J_T3	5	9	K_T3_5	25	U21	HVOUT26	41	38
J_T3	6	7	K_T3_6	26	U21	HVOUT27	40	37
J_T3	7	6	K_T3_7	27	U21	HVOUT28	39	36
J_T3	8	5	K_T3_8	28	U21	HVOUT29	38	35
J_T3	9	4	K_T3_9	29	U21	HVOUT30	37	34
J_T4	0	3	K_T4_0	30	U21	HVOUT31	36	33
J_T4	1	13	K_T4_1	31	U21	HVOUT32	35	32
J_T4	2	12	K_T4_2	32	U22	HVOUT1	22	31
J_T4	3	11	K_T4_3	33	U22	HVOUT2	21	30
J_T4	4	10	K_T4_4	34	U22	HVOUT3	20	29
J_T4	5	9	K_T4_5	35	U22	HVOUT4	19	28
J_T4	6	7	K_T4_6	36	U22	HVOUT5	18	27
J_T4	7	6	K_T4_7	37	U22	HVOUT6	17	26
J_T4	8	5	K_T4_8	38	U22	HVOUT7	16	25
J_T4	9	4	K_T4_9	39	U22	HVOUT8	15	24
J_T5	0	3	K_T5_0	40	U22	HVOUT9	14	23
J_T5	1	13	K_T5_1	41	U22	HVOUT10	13	22
J_T5	2	12	K_T5_2	42	U22	HVOUT11	12	21
J_T5	3	11	K_T5_3	43	U22	HVOUT12	11	20
J_T5	4	10	K_T5_4	44	U22	HVOUT13	10	19
J_T5	5	9	K_T5_5	45	U22	HVOUT14	9	18
J_T5	6	7	K_T5_6	46	U22	HVOUT15	8	17
J_T5	7	6	K_T5_7	47	U22	HVOUT16	7	16
J_T5	8	5	K_T5_8	48	U22	HVOUT17	6	15
J_T5	9	4	K_T5_9	49	U22	HVOUT18	5	14
J_T6	0	3	K_T6_0	50	U22	HVOUT19	4	13
J_T6	1	13	K_T6_1	51	U22	HVOUT20	3	12
J_T6	2	12	K_T6_2	52	U22	HVOUT21	2	11
J_T6	3	11	K_T6_3	53	U22	HVOUT22	1	10
J_T6	4	10	K_T6_4	54	U22	HVOUT23	44	9
J_T6	5	9	K_T6_5	55	U22	HVOUT24	43	8
J_T6	6	7	K_T6_6	56	U22	HVOUT25	42	7
J_T6	7	6	K_T6_7	57	U22	HVOUT26	41	6
J_T6	8	5	K_T6_8	58	U22	HVOUT27	40	5
J_T6	9	4	K_T6_9	59	U22	HVOUT28	39	4
备用	-	-	NC	60	U22	HVOUT29	38	3
备用	-	-	NC	61	U22	HVOUT30	37	2
备用	-	-	NC	62	U22	HVOUT31	36	1
备用	-	-	NC	63	U22	HVOUT32	35	0

说明：通道 = (管位-1)×10 + 数字；CH0..31 → U21，CH32..63 → U22；HVOUT 与 PQFP 引脚按 Microchip DS20005854A Table 2-1（HVOUT1=pin22 … HVOUT22=pin1, HVOUT23=pin44 … HVOUT32=pin35）。

串行位假设：64 bit 帧按 bit0 先发，bit_j = CH(63-j)；实际近端/远端与字节位序须用单通道 walking-one 测试确定，不得用软件数组掩盖镜像。